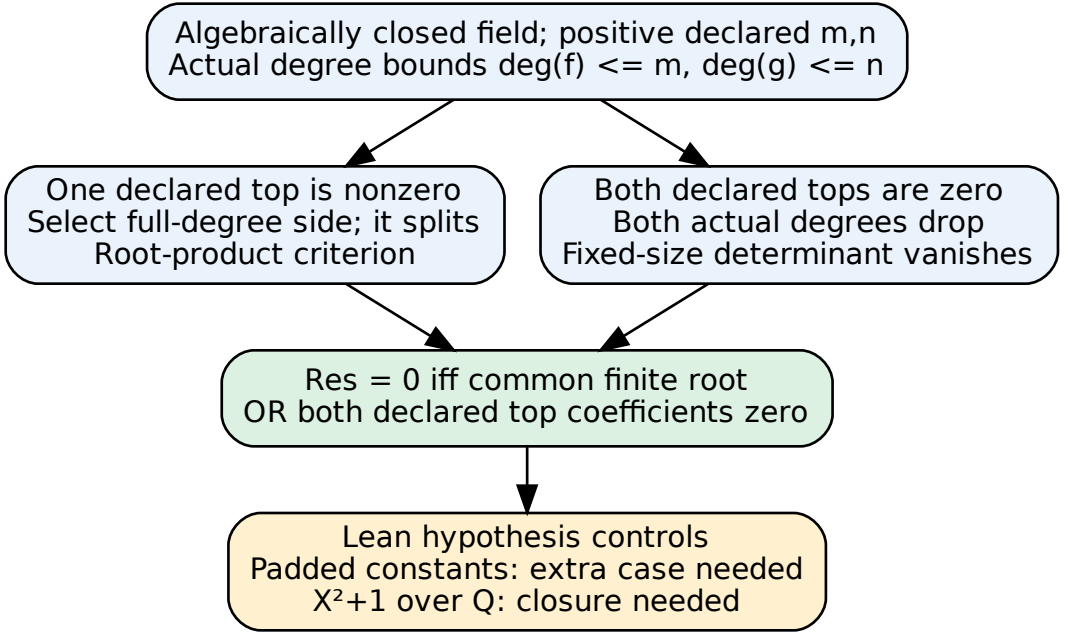
Algebraically closed field; positive declared m,n
Actual degree bounds deg(f) <= m, deg(g) <= n
One declared top is nonzero
Select full-degree side; it splits
Root-product criterion
Both declared tops are zero
Both actual degrees drop
Fixed-size determinant vanishes
Res = 0 iff common finite root
OR both declared top coefficients zero
Lean hypothesis controls
Padded constants: extra case needed
X²+1 over Q: closure needed